BGI HACKATHON 2026

IDEA SUBMISSION

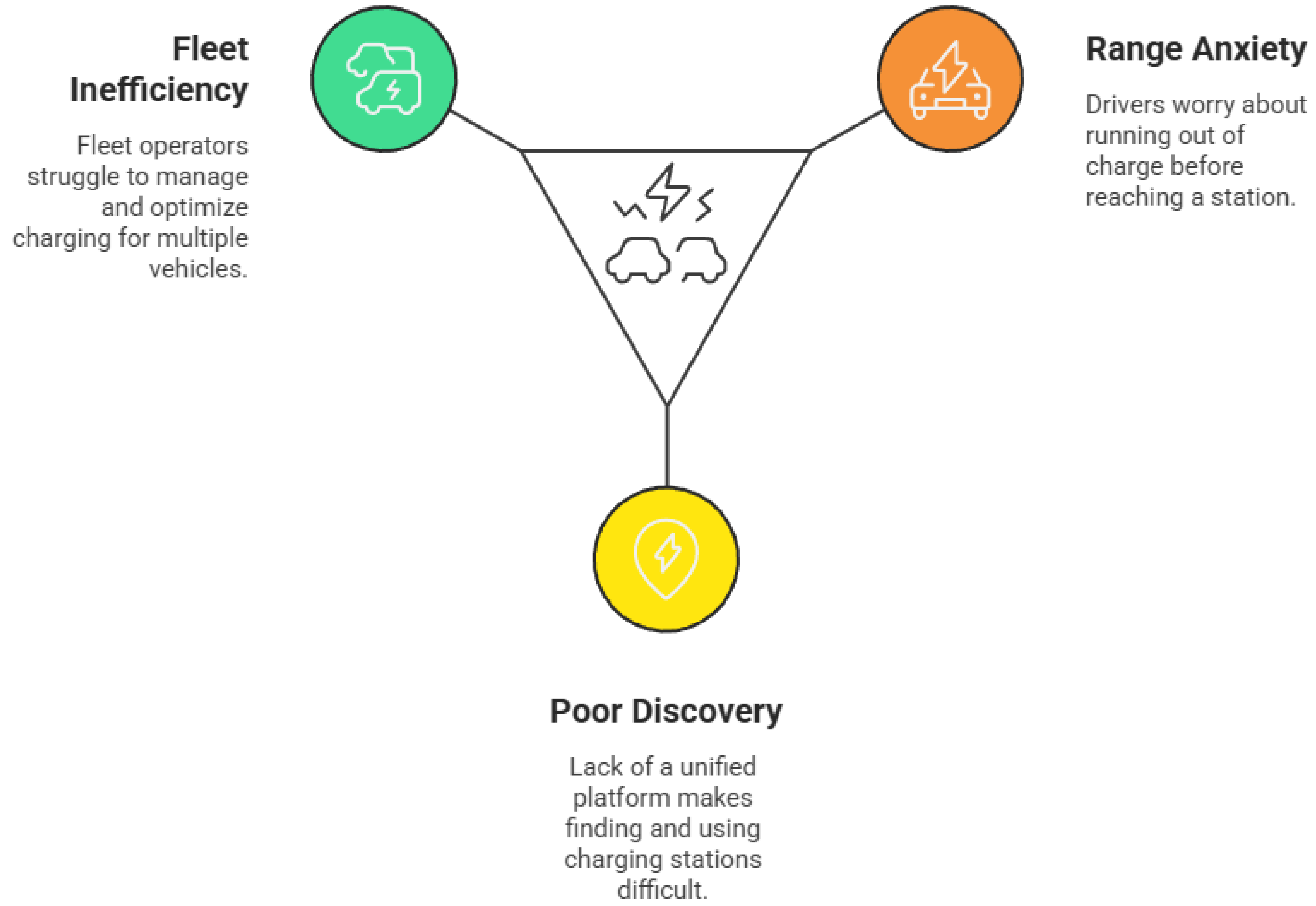
- **Theme-Smart Mobility, EVs & Logistics**

Your paragraph text

- **Problem Statement ID –BT2P2**
- **Problem Statement Title-EV Charging Station Locator & Slot Booking System**
- **Problem Statement Category- Software(primary)/Hardware(optional)**
- **Team ID-833**
- **Team Name (Registered on portal) -CivicShield**

SWIFT CHARGE — AI-Powered Smart EV Charging Intelligence Platform

EV Charging Challenges



OLUTION

Find — AI Demand Map

Real-time map with predictive occupancy heatmap. Know which station will free before you drive

Book — Smart Reservations

Reserve slots up to 4 hours ahead. UPI/Razorpay payment. Dynamic pricing. tap cancel

Drive — Route-Aware Nav

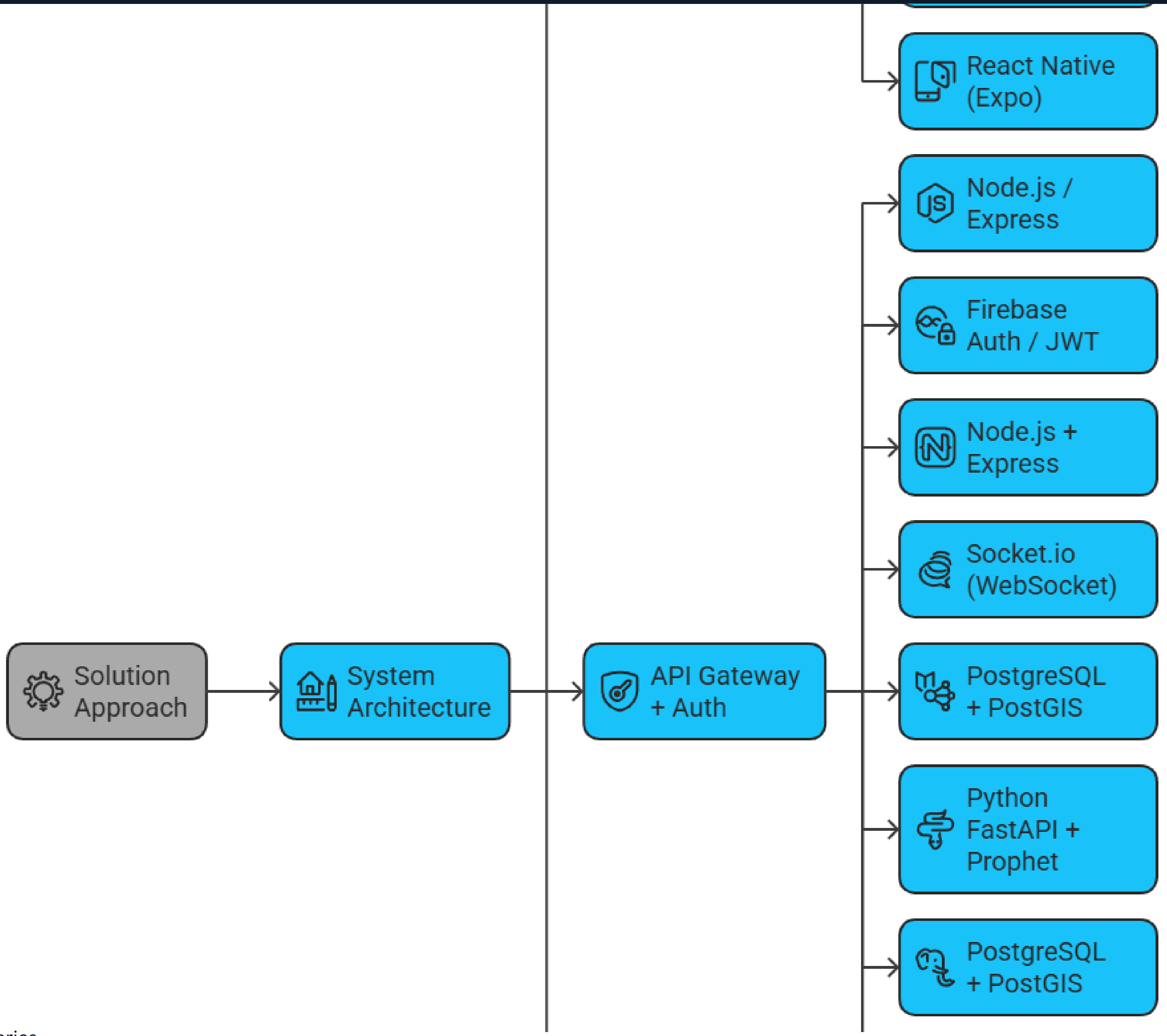
EV-smart trip planner: computes optimal charging stops across full journey live slot booking

SOLUTION APPROACH

YSTEM ARCHITECTURE

User Interfaces	Mobile App (React Native)	Web App (React+Vite)
API Gateway + Auth	Node.js / Express	Firebase Auth / JWT
Microservices	Location Svc	Booking Svc
Data Layer	PostgreSQL + PostGIS	Redis (slot locks)
IoT Layer (Optional)	MQTT Broker	NodeMCU Sensors

IOS



KEY TECHNICAL DECISIONS

- Redis slot-lock (3-min optimistic lock prevents double-booking)
- Prophet ML for demand time-series forecasting
- PostGIS ST_DWithin for fast geo-radius queries
- Socket.io rooms for per-station real-time updates

FEASIBILITY AND VIABILITY

CHALLENGES & MITIGATION STRATEGIES

Real-time data accuracy

Risk: Stale slot data leads to user frustration and failed bookings

Fix: WebSocket push from IoT sensors + Redis TTL-based cache invalidation ensures <2s update latency

Concurrent booking conflicts

Risk: Two users booking the same slot simultaneously causes double-booking

Fix: Optimistic lock in Redis (3-min TTL) + atomic DB transaction with conflict rollback

Scalability to 1000+ stations

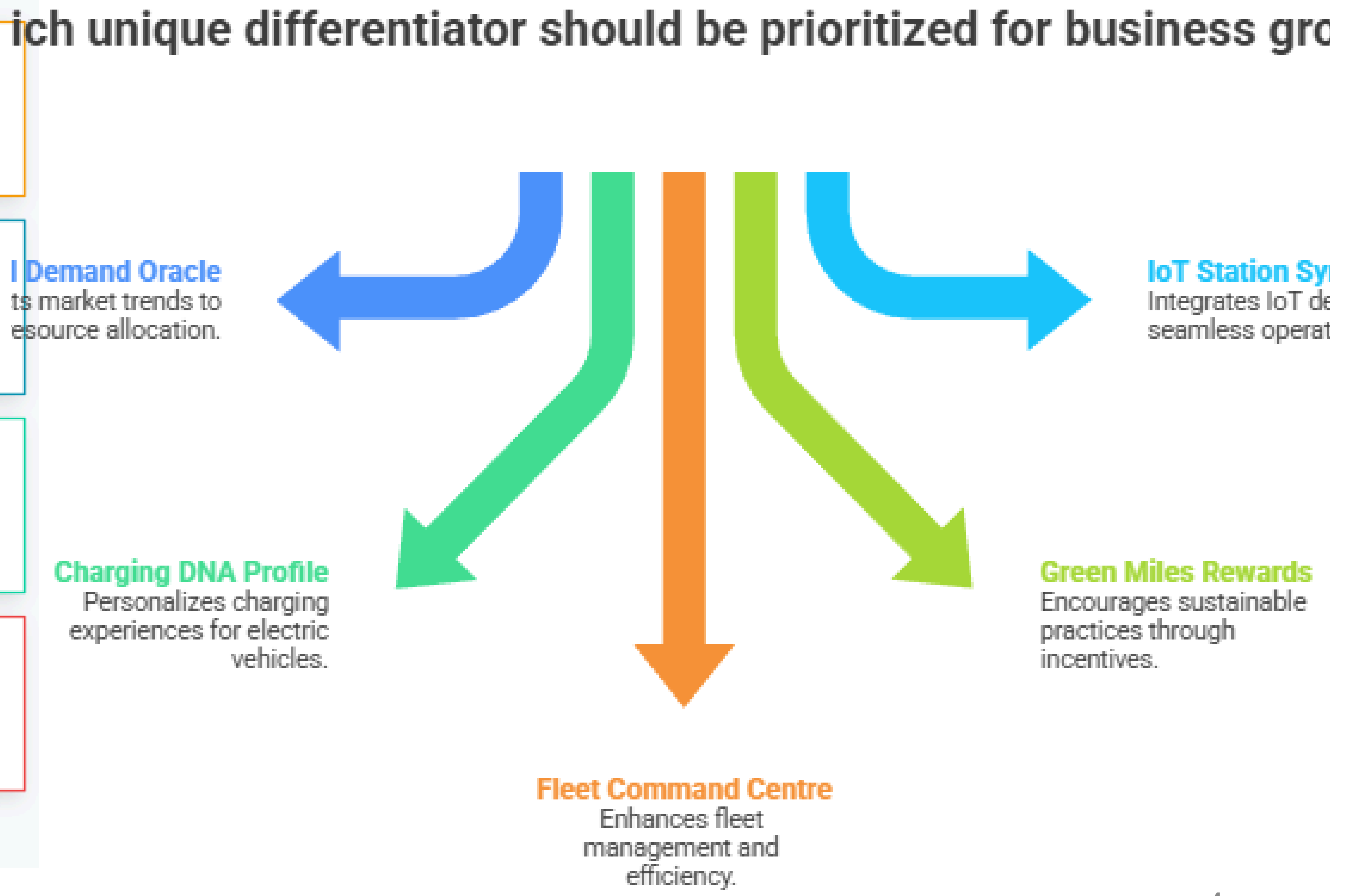
Risk: Geospatial queries slow down as station count grows

Fix: PostGIS spatial index (GIST) + horizontal read replicas. Sub-50ms query on 10k+ stations

Payment security & fraud

Risk: Financial transactions require PCI-DSS compliance

Fix: Razorpay handles PCI compliance. No card data stored. Server-side payment verification only



IMPACT AND BENEFITS

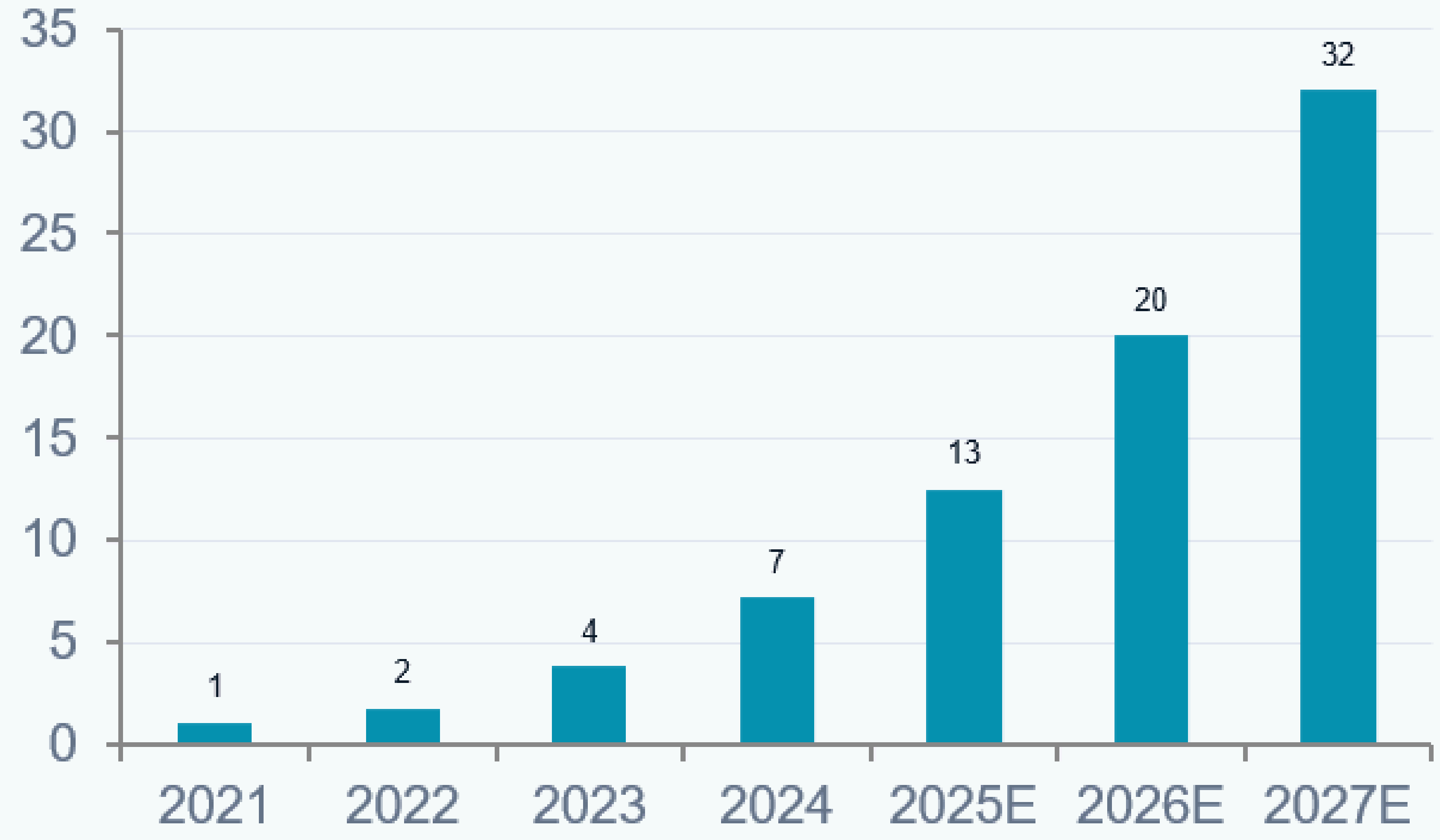
65%
Wait time reduction
via smart booking

40%
Station utilisation
via AI scheduling

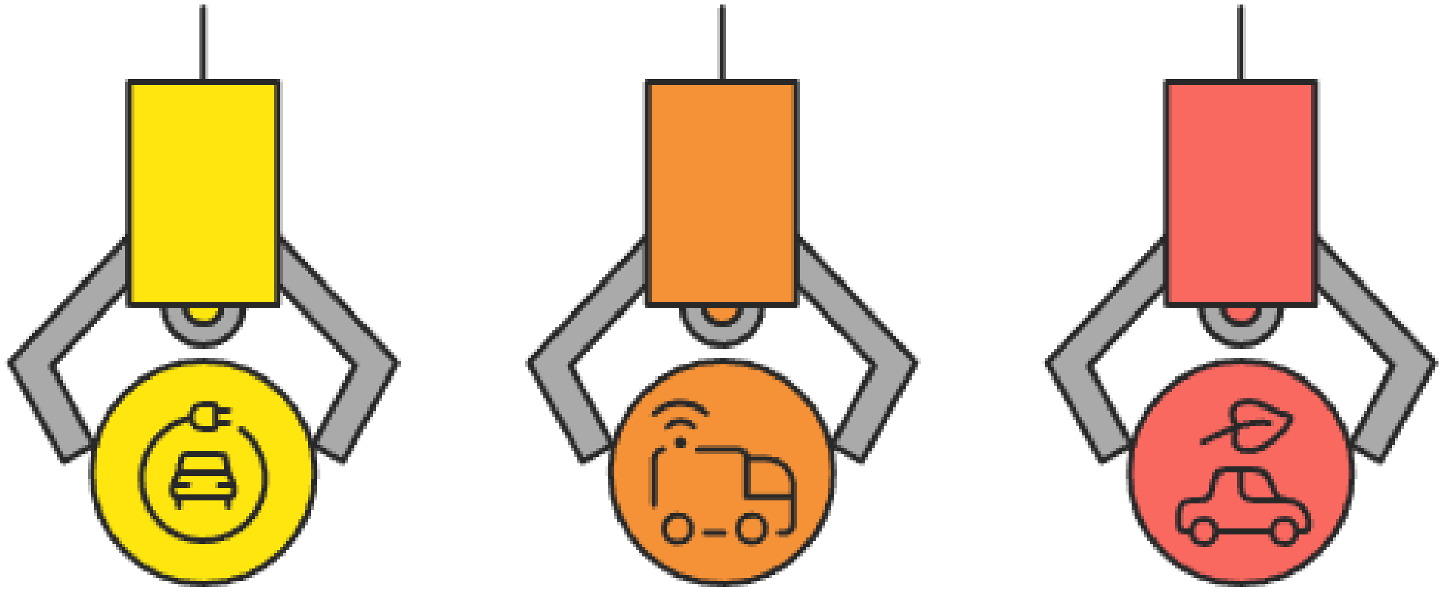
1.8T
CO_2 saved / user / yr
vs petrol vehicle

30%
Fleet cost saving
via optimised charging

INDIA EV ADOPTION FORECAST (Millions)



Stakeholder Benefits



Individual EV Users

No more range anxiety or guesswork, book charging slots from your phone in 30 seconds.

Fleet Operators / Businesses

Centralised dashboard for all vehicles, bulk booking and schedule management.

Environment & Society

Reduces EV range anxiety, accelerating EV adoption and supporting India's net-zero roadmap.

Research and References

Solution Component Effort

Demand prediction framework and station IoT layer are key components.

Market Research

NITI Aayog and BNEF reports provide insights into India's EV sector. SIAM reports on EV sales growth.

Technology Standards

WebSocket, Firebase, Razorpay, OpenChargeMap API, V2G, Prophet, and MQTT are used.

Reseach and References
NITI Aayog EV Report (PDF)
BloombergNEF EV Outlook Report
SIAM EV Industry Data (India)

TECHNOLOGY STANDARDS USED

PostGIS
Spatial extension for PostgreSQL — geo-radius queries

RFC 6455
WebSocket protocol standard — real-time updates

OpenID Connect
Auth protocol via Firebase — secure token flow

PCI-DSS
Payment security standard — via Razorpay

OCPP 2.0.1
Open Charge Point Protocol — station integration

ISO 15118